# The engineers who revolutionised Formula 1

*From Cooper to Newey: who really changed the technical grammar of the sport?*

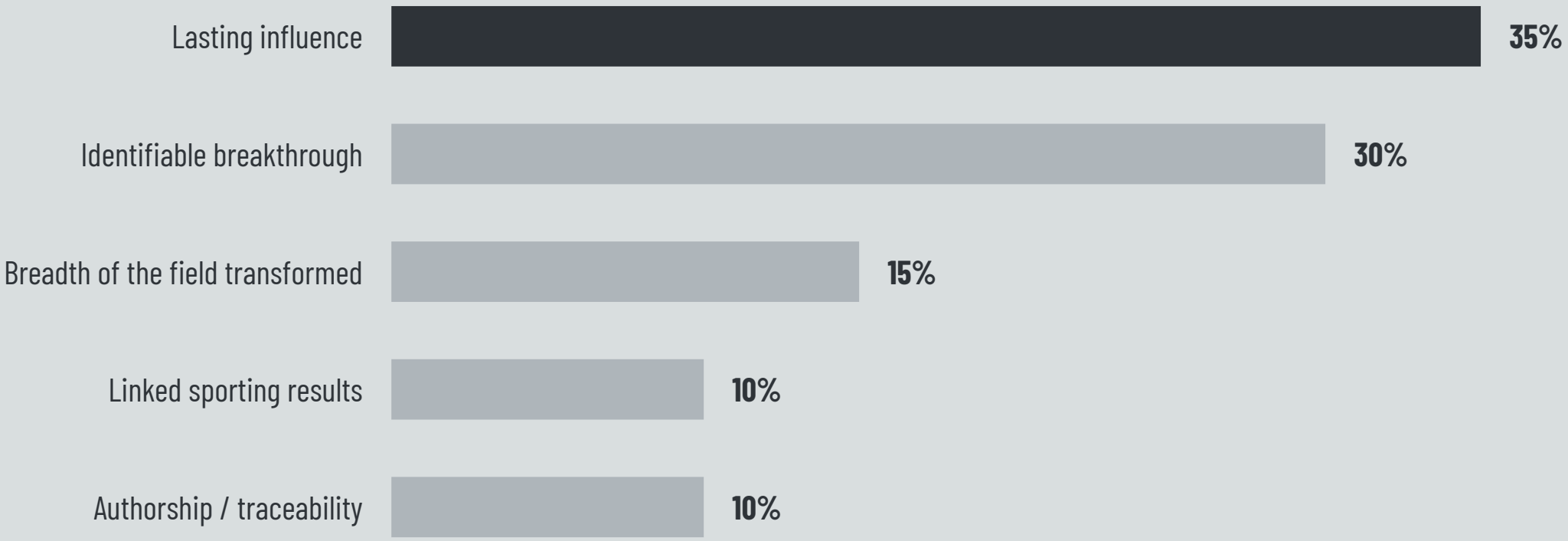


# “Who won the most?” is not the right question

| | | |
|---|---|---|
| **Titles** | ≠ | **Innovation** |
| **Innovation** | ≠ | **Invention** |
| **Invention** | ≠ | **Diffusion** |
| **Technical direction** | ≠ | **Personal authorship** |

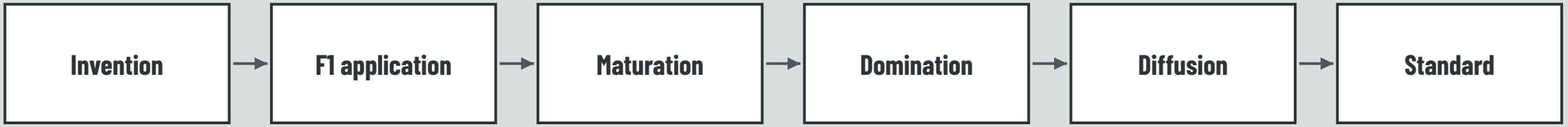
*The ranking measures the capacity to shift the technical frontier for good.*

# Five criteria, led by lasting influence




*The scores are a comparative instrument, not historical facts.*

# An innovation has several authors, at different stages




**Examples**

| | | | |
|---|---|---|---|
| Ground effect | Cooper rear engine | Cosworth DFV | Williams active suspension |

# Twenty engineers, four tiers

## Tier A
Paradigm revolutionaries


Colin Chapman


John Barnard

JC John Cooper

KD Keith Duckworth


Adrian Newey

## Tier B
Major systemic transformers

BD Bernard Dudot


Gordon Murray


Mauro Forghieri

PW Peter Wright

FD Frank Dernie

## Tier C
Structuring innovators

PH Patrick Head


Ross Brawn

AC Andy Cowell

JM Jean-Claude Migeot


Tony Rudd

## Tier D
More localised or more shared impact


Mike Costin


Vittorio Jano

OM Owen Maddock


Paddy Lowe


Rory Byrne

*Photos: credits on slide 22*

# Five questions that changed F1

| | | | |
|---|---|---|---|
| 1 | JC | **Cooper** | Where should the engine go? |
| 2 |  | **Chapman** | How should the car be built? |
| 3 | KD | **Duckworth** | How should the engine be integrated? |
| 4 |  | **Barnard** | With which materials and interfaces? |
| 5 |  | **Newey** | How can the whole aerodynamic system be optimised? |

*Photos: credits on slide 22*

Colin Chapman

# Three breakthroughs, one philosophy: architecture + lightness + aero

## Lotus 25

Aluminium monocoque

## Lotus 49

Engine as a stressed member

## Lotus 78 / 79

Ground effect programme



**Shared attribution:** Wright / Rudd for ground effect.

| Rank | Tier | Score | Confidence |
|---|---|---|---|
| 1 | A | 96.0 | High |

*Sources: Lotus — Type 25 · Lotus — Lotus Story · F1 — Tech Tuesday: The Lotus 79 · Photo: Joost Evers / Anefo (Nationaal Archief) (CC0, Wikimedia Commons), cropped and converted to black and white*

John Barnard

# Two independent innovations that became universal

**1981**

**McLaren MP4/1**

Full carbon-fibre monocoque

**1989**

**Ferrari 640**

Semi-automatic gearbox, paddles on the wheel



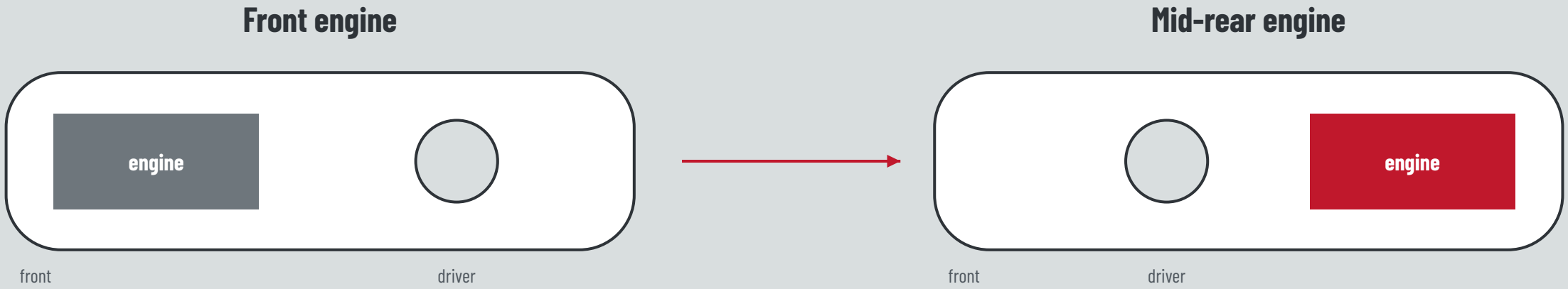
*The clearest personal authorship: Chapman's main challenger.*

| Rank | Tier | Score | Confidence |
|---|---|---|---|
| 2 | A | 95.0 | Very high |

*Sources: McLaren — MP4/1 · F1 — John Barnard podcast · F1 — How F1 looked in 1990 · Photo: B3 Technologies (CC BY-SA 3.0, Wikimedia Commons), cropped and converted to black and white*

John Cooper

# A topological revolution of the single-seater



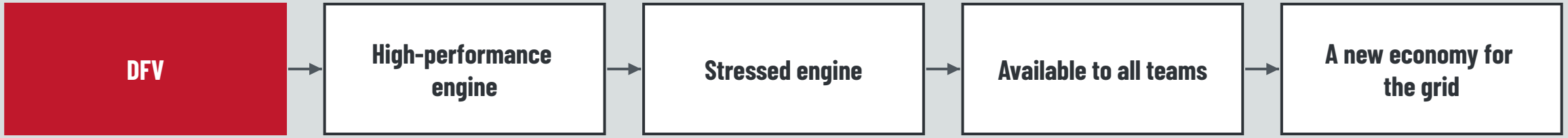

**Cooper T51, 1959 and 1960 titles**
Chief designer: Owen Maddock

| Rank | Tier | Score | Confidence |
|---|---|---|---|
| 3 | A | 93.5 | High |

*Sources: Motor Sport — Cooper T51 · Goodwood / Doug Nye — Cooper · F1 — Most influential technical innovator*

Keith Duckworth

# A technical and economic revolution at once




**155** world championship Grand Prix wins, 1967-1983

**Cosworth = Duckworth + Costin + an organisation**

| Rank | Tier | Score | Confidence |
|---|---|---|---|
| 4 | A | 92.5 | High |

*Sources: F1 — Most influential technical innovator · Motor Sport — The Lotus 49 (1967)*

# Newey's genius is repetition, not a single stroke

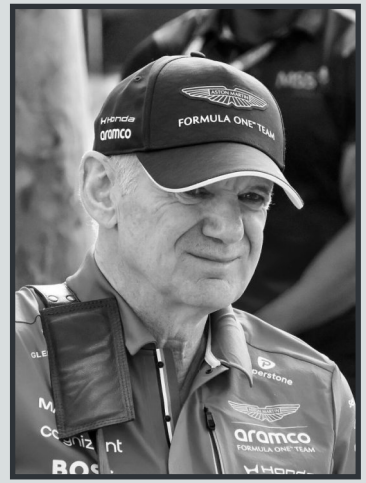
**March / Leyton House** — 1988-1990

**Williams** — 1991-1996

**McLaren** — 1997-2005

**Red Bull** — 2006-2024

**Aston Martin** — since 2025




Different regulations, the same skill: mastery of the aerodynamic system.

| Rank | Tier | Score | Confidence |
|---|---|---|---|
| 5 | A | 91.5 | Very high |

*Sources: F1 — Most influential technical innovator · F1 — Newey's Red Bull exit · F1 — How F1 looked in 1990 · Photo: Yu Chu Chin (CC BY-SA 4.0, Wikimedia Commons), cropped and converted to black and white*

The real debate for first place

# Why Chapman ahead of Barnard?



| Criterion | Chapman | Barnard |
|---|---|---|
| Number of breakthroughs | ++ | + |
| Direct authorship | + | ++ |
| Architectural reach | ++ | ++ |
| Diffusion | ++ | ++ |
| Repetition | ++ | + |



*Chapman wins on historical breadth; Barnard wins on personal traceability.*

*Photo: Joost Evers / Anefo (Nationaal Archief) (CC0, Wikimedia Commons), cropped and converted to black and white · Photo: B3 Technologies (CC BY-SA 3.0, Wikimedia Commons), cropped and converted to black and white*

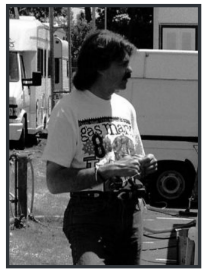
# Tier B: five systemic transformers

BD

**6**

**Dudot**

Power



**7**

**Murray**

Radical architecture



**8**

**Forghieri**

Total engineering

PW

**9**

**Wright**

Ground effect

FD

**10**

**Dernie**

A silent digital revolution

*Compared with Tier A: a more collective attribution, a more specialised reach or uneven diffusion.*

*Sources: Renault — 30 ans de Formule 1 · F1 — Most Influential Person: final 4 · F1 — Ferrari 312T · F1 — Tech Tuesday: The Lotus 79 · Motor Sport — Lunch with Frank Dernie · Photos: credits on slide 22*

# The engine revolution: turbo, pneumatic valves, V10




Adopted by the whole grid

The general solution in the 1990s

67° V10 under Dudot's direction

Renault Sport is an organisation: Dudot is not a lone inventor.

| Rank | Tier | Score | Confidence |
|---|---|---|---|
| 6 | B | 90.0 | High |

*Sources: Renault — 30 ans de Formule 1 · Motor Sport — Renault Returns*

# The silent revolutions

## Frank Dernie

- CAD / CAE
- Data acquisition
- Wind tunnel
- Active suspension

FD

## Ross Brawn

- Methods
- Organisation
- Early programmes
- Integration



*A technical revolution can transform the process without showing on the bodywork.*

*Sources: Motor Sport — Lunch with Frank Dernie · F1 — Ross Brawn favourite car designs · Photos: credits on slide 22*

# Ground effect: restoring the causal chain




Lotus 78: drawn by Ralph Bellamy, suspension by Martin Ogilvie, development with Tony Southgate.

*Sources: F1 — Tech Tuesday: The Lotus 79*

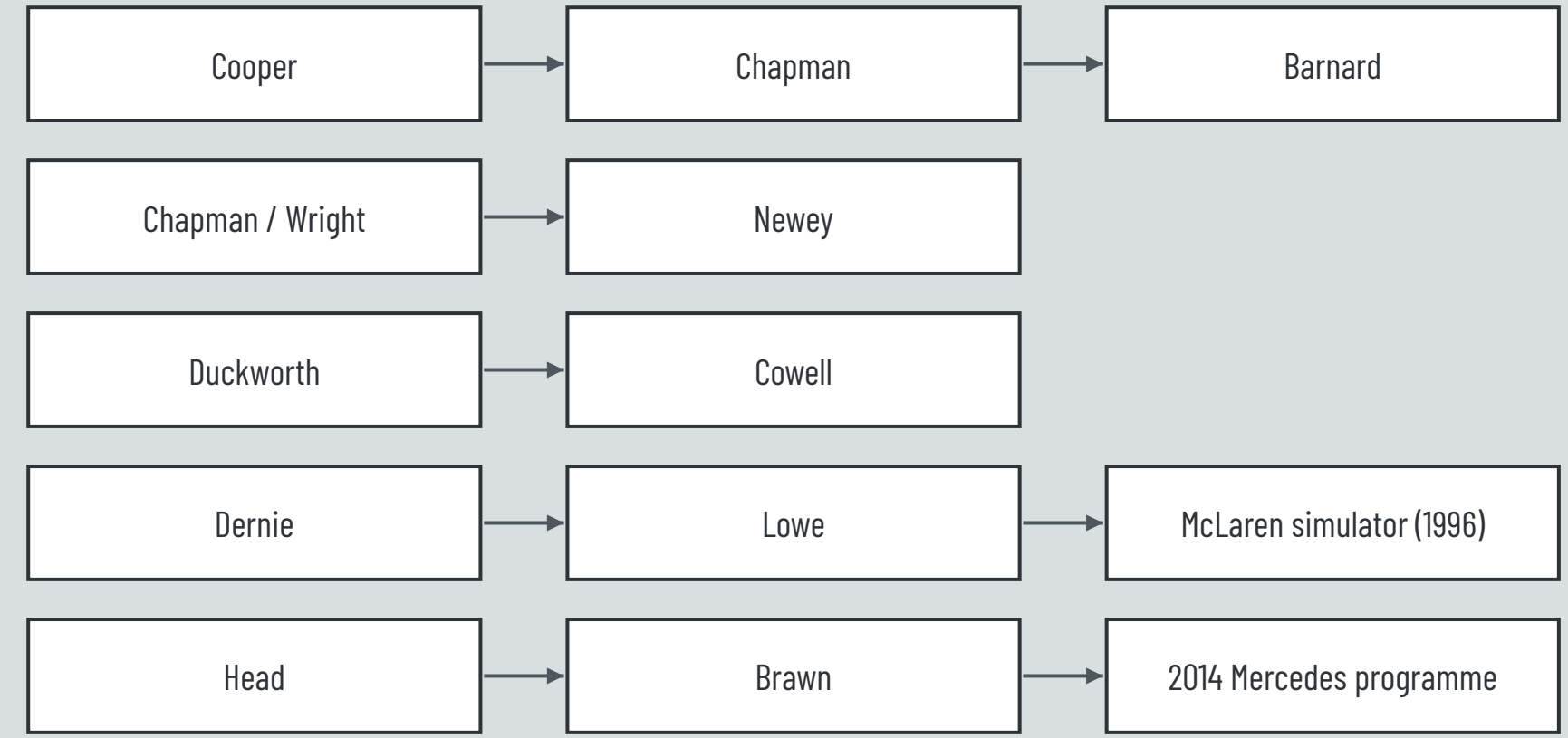
Ranks 11 to 20

# Tiers C and D: ten major contributions

| Rank | | Engineer | Field | Reason for the ranking | Limit |
|---|---|---|---|---|---|
| 11 | PH | **Patrick Head** | Systems engineering | Williams technical organisation | Organisation rather than a single breakthrough |
| 12 |  | **Ross Brawn** | Method | Early development, integration | A revolution in process |
| 13 | AC | **Andy Cowell** | Power unit | Mercedes 2014 hybrid | Hybridisation imposed by the regulations |
| 14 | JM | **Jean-Claude Migeot** | Aerodynamics | Raised nose, Tyrrell 019 | Quickly absorbed by general evolution |
| 15 |  | **Tony Rudd** | Aerodynamics | Origins of ground effect | Contribution hard to isolate |
| 16 |  | **Mike Costin** | Engine | Co-founder of Cosworth | DFV authorship less individual |
| 17 |  | **Vittorio Jano** | Architecture | Lancia D50 | Work mostly before F1 |
| 18 | OM | **Owen Maddock** | Chassis | Chief designer, 1959-1960 titles | Credit shared with John Cooper |
| 19 |  | **Paddy Lowe** | Electronics and software | Active suspension software | A role in a collective chain |
| 20 |  | **Rory Byrne** | Chassis and aerodynamic integration | Benetton, Ferrari | No single paradigmatic breakthrough |

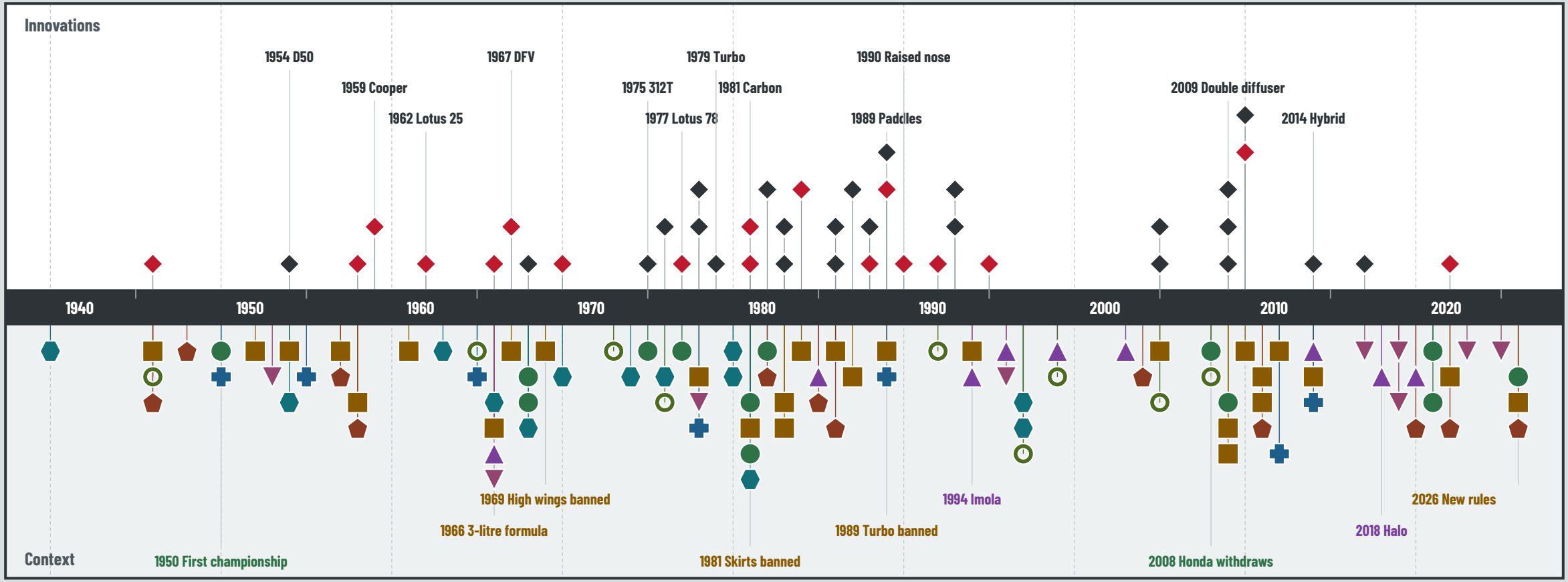
*Sources: F1 — Williams FW15C · F1 — Ross Brawn favourite car designs · F1 — How F1 looked in 1990 · Stellantis Heritage — Lancia D50 · Motor Sport — Cooper T51 · Photos: credits on slide 22*

# The great technical lineages




*This diagram shows intellectual lineages, not master-pupil relationships.*

# Eighty years of breakthroughs, and their context

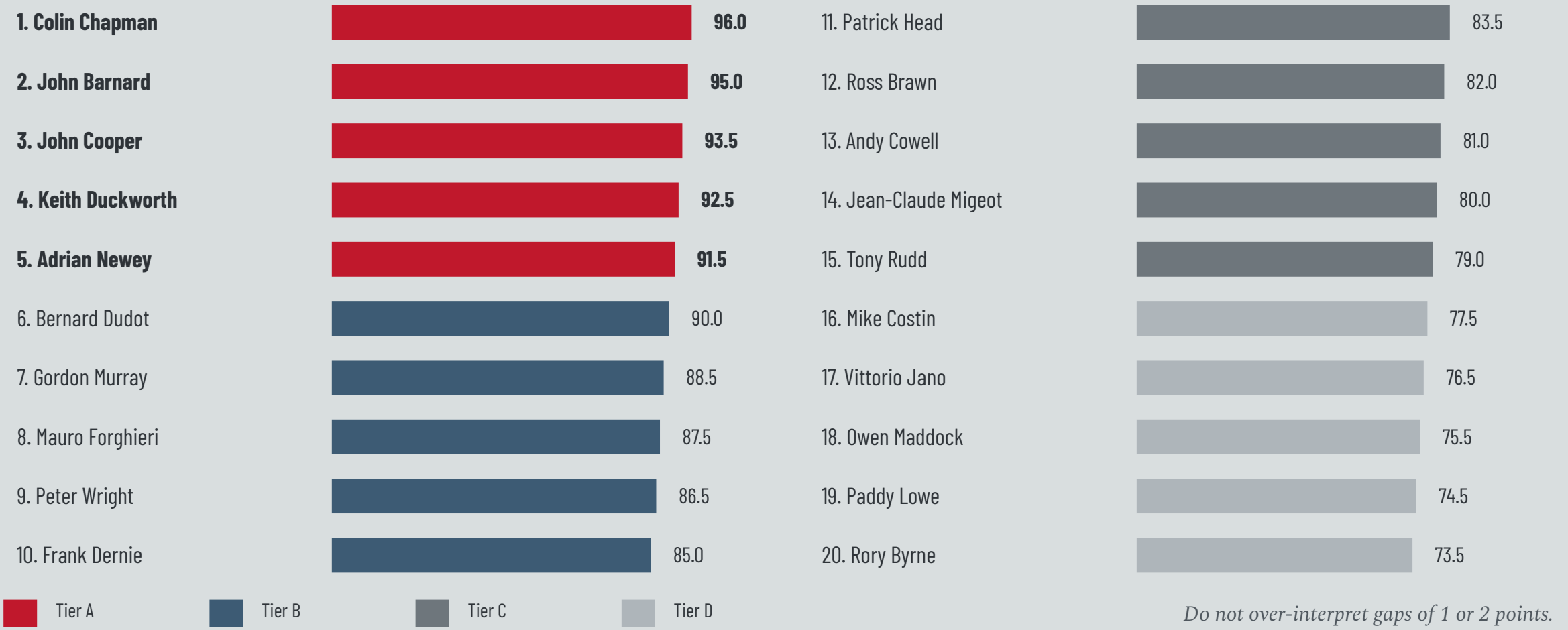



innovation, Tier A · innovation · Regulations · Safety · Economics · Technology · Geopolitics · World economy · Other sports · Society

*Interactive timeline: s-geffroy.github.io/f1-eng/en/timeline*

Scores out of 100

# The final ranking




*Do not over-interpret gaps of 1 or 2 points.*

# Revolutionising F1 is not just about building the fastest car: it is about forcing others to change the way they build an F1 car.

| | |
|---|---|
| **Chapman** | *multiple breakthroughs* |
| **Barnard** | *the clearest authorship* |
| **Newey** | *the most durable adaptation* |

Consulted on 9 August 2026

# Main sources


- F1 — Most influential technical innovator
- F1 — The 75 best… in F1 history
- F1 — Tech Tuesday: The Lotus 79
- F1 — John Barnard podcast
- F1 — Williams FW15C
- Lotus — Type 25
- McLaren — MP4/1
- Renault — 30 ans de Formule 1
- Motor Sport — Cooper T51
- Motor Sport — Lunch with Frank Dernie


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*Full bibliography, with addresses: speaker notes (.pptx) and s-geffroy.github.io/f1-eng/en*